Illustrative Mathematics 6–8 大纲与海报计划

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海报粒度：和 K–5 一样，每个 Section 一张。全套 119 个 Section（六年级 43、七年级 36、八年级 40），不超过约 250 张，所以按 Section 出海报。每个单元最后的「Let’ s Put It to Work（综合运用）」部分官方没有列 Section Goals，大纲里改为摘录该部分的说明。

本地化：英制单位改为公制，美元改为人民币，美国特有情境（销售税、小费等）改为本地情境，涉及的 Section 下面有标注。海报里为说明而举的数字都标了「示例」；动手活动加了安全提示。

来源

- Grade 6：课程首页 · Scope and Sequence · 单元页如 Unit 1
- Grade 7：课程首页 · Scope and Sequence · 单元页如 Unit 1
- Grade 8：课程首页 · Scope and Sequence · 单元页如 Unit 1
- 总页：IM 6–8 Math（accessim.org） · IM 6–8 Math 介绍

六年级（Grade 6）· 9 个单元 · 43 个 Section

第1单元 面积和表面积（Unit 1: Area and Surface Area）

A部分 用推理求面积（Section A: Reasoning to Find Area）

- 学习目标：Find the area of a two-dimensional region with straight boundaries by decomposing, rearranging, subtracting, or enclosing shapes, and explain the solution method.
- 课：6.1.1 Tiling the Plane；6.1.2 Finding Area by Decomposing and Rearranging；6.1.3 Reasoning to Find Area
- 海报要点：分割、重组、割补，图形完全重合面积相等，面积可以相加

B部分 平行四边形（Section B: Parallelograms）

- 学习目标：Apply the formula for area of a parallelogram to find the area, the length of a base, or the height.
- 学习目标：Understand bases and heights in a parallelogram and recognize base-height pairs to use to calculate its area.
- 课：6.1.4 Parallelograms；6.1.5 Bases and Heights of Parallelograms；6.1.6 Area of Parallelograms
- 海报要点：底和对应的高，面积 = 底 × 高

C部分 三角形和其他多边形（Section C: Triangles and Other Polygons）

- 学习目标：Find the area of a polygon by decomposing it into parallelograms and triangles.
- 学习目标：Understand bases and heights in a triangle and recognize base-height pairs to use to find the area of a triangle.
- 学习目标：Understand why the process of finding the area of a triangle can be abstracted as 1/2 · b · h (or equivalent) and apply the formula to find the area of a triangle.
- 课：6.1.7 From Parallelograms to Triangles；6.1.8 Area of Triangles；6.1.9 Formula for the Area of a Triangle；6.1.10 Bases and Heights of Triangles；6.1.11 Polygons
- 海报要点：三角形是平行四边形的一半，面积 = ½ × 底 × 高；多边形分割成三角形

D部分 表面积（Section D: Surface Area）

- 学习目标：Identify or create a net that represents a prism or pyramid.
- 学习目标：Use a net to calculate the surface area of a prism or pyramid and explain the solution method.
- 课：6.1.12 What Is Surface Area?；6.1.13 Polyhedra；6.1.14 Nets and Surface Area；6.1.15 More Nets, More Surface Area；6.1.16 Distinguishing Between Surface Area and Volume (Optional)
- 海报要点：多面体、展开图，用展开图求棱柱、棱锥的表面积

E部分 正方形和正方体（Section E: Squares and Cubes）

- 学习目标：Interpret and write expressions with exponents 2 and 3 to represent the area of a square or the volume of a cube.
- 学习目标：Write expressions, with or without exponents, to represent the surface area of a given cube.
- 课：6.1.17 Squares and Cubes；6.1.18 Surface Area of a Cube
- 海报要点：用指数 ^{2} 和 ^{3} 表示正方形面积、正方体体积；正方体表面积 $6s^2$

F部分 综合运用：帐篷（Section F: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions. The lesson in this section is optional because it offers additional opportuniti…
- 课：6.1.19 All about Tents (Optional)
- 海报要点：估算、计算做帐篷需要多少布料
- 本地化 / 安全：原书帐篷尺寸用英尺，本海报用米，数字是示例。

第2单元 认识比（Unit 2: Introducing Ratios）

A部分 什么是比（Section A: What Are Ratios?）

- 学习目标：Create and interpret discrete diagrams that represent situations involving ratios.
- 学习目标：Create and interpret sentences that describe ratios.
- 课：6.2.1 Introducing Ratios and Ratio Language；6.2.2 Representing Ratios with Diagrams
- 海报要点：用比的语言描述两个量，用离散图表示比

B部分 等值比（Section B: Equivalent Ratios）

- 学习目标：Explain equivalent ratios in terms of the quantities in a recipe being multiplied by the same number to create a different size batch of something with the same taste or same shade.
- 学习目标：Generate equivalent ratios and justify that they are equivalent.
- 课：6.2.3 Recipes；6.2.4 Color Mixtures；6.2.5 Defining Equivalent Ratios
- 海报要点：食谱、调色：各部分同乘一个数，味道、颜色不变

C部分 表示等值比（Section C: Representing Equivalent Ratios）

- 学习目标：Given prices and quantities or distances and times, calculate the price for 1 object or the distance traveled in 1 unit of time. Express each using the word “per.”
- 学习目标：Justify whether two situations do or do not happen at the same rate, by determining if the ratios in the situations are equivalent.
- 学习目标：Use a double number line diagram to represent and find equivalent ratios.
- 课：6.2.6 Introducing Double Number Line Diagrams；6.2.7 Creating Double Number Line Diagrams；6.2.8 How Much for One?；6.2.9 Constant Speed；6.2.10 Comparing Situations by Examining Ratios
- 海报要点：双数线；每 1 份是多少（单价、速度）；比较两种情况是否同样快

D部分 解比和比率问题（Section D: Solving Ratio and Rate Problems）

- 学习目标：Choose multipliers strategically when solving multi-step problems involving equivalent ratios.
- 学习目标：Use a table of equivalent ratios to solve problems.
- 课：6.2.11 Representing Ratios with Tables；6.2.12 Navigating a Table of Equivalent Ratios；6.2.13 Tables and Double Number Line Diagrams；6.2.14 Solving Equivalent Ratio Problems
- 海报要点：比率表，策略性地选择乘数

E部分 部分与整体的比（Section E: Part-Part-Whole Ratios）

- 学习目标：Choose representations and solution methods to reason about ratios and sums of quantities.

- 学习目标：Use diagrams or other strategies to solve problems involving ratios and the total amount.
- 课：6.2.15 Part-Part-Whole Ratios；6.2.16 Solving More Ratio Problems
- 海报要点：部分 : 部分 : 整体，已知总量求各部分

F部分 综合运用：费米问题（Section F: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：In the final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：6.2.17 A Fermi Problem
- 海报要点：用比和估算解决费米问题

第3单元 单位比率和百分数（Unit 3: Unit Rates and Percentages）

A部分 测量单位与换算（Section A: Units of Measurement and Unit Conversion）

- 学习目标：Recognize that when we measure things in two different units, the pairs of measurements are equivalent ratios. Apply this understanding to convert a measurement from one unit to another unit.
- 学习目标：Use a “rate per 1” to solve problems involving unit conversion.
- 课：6.3.1 Anchoring Units of Measurement；6.3.2 Measuring with Different-Size Units (Optional)；6.3.3 Converting Units
- 海报要点：同一个量用两种单位测，测量值成等值比；用「每 1」换算
- 本地化 / 安全：原书同时使用英制单位（英寸、英尺、杯、磅等）和公制单位，本海报只用公制单位换算。

B部分 比率（Section B: Rates）

- 学习目标：Apply reasoning about ratios and unit rates to solve problems and explain the solution methods.
- 学习目标：Calculate the two unit rates associated with a ratio and interpret them in the context of a situation.
- 课：6.3.4 Comparing Speeds and Prices；6.3.5 Interpreting Rates；6.3.6 Equivalent Ratios Have the Same Unit Rates；6.3.7 More Rate Comparisons；6.3.8 Solving Rate Problems；6.3.9 More about Constant Speed (Optional)
- 海报要点：一个比对应两个单位比率；比较速度和价格
- 本地化 / 安全：原书价格用美元，本海报改用人民币。

C部分 百分数（Section C: Percentages）

- 学习目标：Choose a strategy to solve problems involving percentages and explain the solution method.
- 学习目标：Create tape diagrams, double number line diagrams, or tables to represent situations involving percentages.
- 课：6.3.10 What Are Percentages?；6.3.11 Representing Percentages with Double Number Line Diagrams；6.3.12 Representing Percentages in Different Ways；6.3.13 Benchmark Percentages；6.3.14 Solving Percentage Problems；6.3.15 Finding This Percent of That；6.3.16 Finding the Percentage (Optional)
- 海报要点：百分数是每 100 中的几；用双数线、线段图、表格解百分数问题

D部分 综合运用：粉刷房间（Section D: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：In the final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：6.3.17 Painting a Room
- 海报要点：算墙面面积、油漆用量和费用
- 本地化 / 安全：原书单位为英尺、加仑、美元，本海报改用米、升、人民币，数字是示例。

第4单元 分数除法（Unit 4: Dividing Fractions）

A部分 理解除法（Section A: Making Sense of Division）

- 学习目标：Create a diagram and a multiplication or division equation to represent the relationship in situations involving equal-size groups, and coordinate these representations.
- 学习目标：Interpret a division expression in two ways: as an answer to a “How many groups?” question or to a “How many in each group?” question.
- 课：6.4.1 Size of Divisor and Size of Quotient；6.4.2 Meanings of Division；6.4.3 Interpreting Division Situations (Optional)
- 海报要点：除法的两种意思：有几组？每组多少？

B部分 分数除法的含义（Section B: Meanings of Fraction Division）

- 学习目标：Create a diagram, a multiplication equation, and a division equation to represent a problem that involves a fractional divisor and that asks “How many groups?” or “How much in one group?”
- 学习目标：Solve problems involving division of fractions by using diagrams, writing equations, or reasoning about the relationship between multiplication and division. Explain the solution methods.
- 课：6.4.4 How Many Groups? (Part 1) (Optional)；6.4.5 How Many Groups? (Part 2)；6.4.6 Using Diagrams to Find the Number of Groups；6.4.7 What Fraction of a Group?；6.4.8 How Much in Each Group? (Part 1)；6.4.9 How Much in Each Group? (Part 2) (Optional)
- 海报要点：用图和乘法、除法算式表示除数是分数的问题

C部分 分数除法的算法（Section C: Algorithm for Fraction Division）

- 学习目标：Generalize a process for dividing a number n by a fraction a/b as n ・ b ・ 1/a, n ・ b ÷ a, or n ・ b/a.
- 学习目标：Use the fraction division algorithm to calculate quotients.
- 课：6.4.10 Dividing by Unit and Non-Unit Fractions；6.4.11 Using an Algorithm to Divide Fractions
- 海报要点：除以 a/b，等于乘 b/a

D部分 长度面积体积中的分数（Section D: Fractions in Lengths, Areas, and Volumes）

- 学习目标：Determine the volume of a rectangular prism by packing it with unit cubes with appropriate unit-fraction edge lengths and by multiplying the edge lengths.
- 学习目标：Solve problems about length comparison, the side lengths and area of a rectangle, and the edge lengths and volume of a rectangular prism using fraction division.
- 课：6.4.12 Fractional Lengths；6.4.13 Rectangles with Fractional Side Lengths；6.4.14 Fractional Lengths in Triangles and Prisms；6.4.15 Volume of Prisms
- 海报要点：分数边长的长方形面积、分数棱长的长方体体积

E部分 综合运用（Section E: Let’s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions. The first lesson in this section is optional because it consolidates the concep…
- 课：6.4.16 Solving Problems Involving Fractions (Optional)；6.4.17 Fitting Boxes into Boxes
- 海报要点：分数四则问题；用分数算装箱和运费
- 本地化 / 安全：原书运费用美元、长度用英寸，本海报改用人民币和厘米，数字是示例。

第5单元 十进制运算（Unit 5: Arithmetic in Base Ten）

A部分 小数的加减（Section A: Exploring, Adding, and Subtracting Decimals）

- 学习目标：Explain the sum or difference of two decimals in terms of combining like base-ten units and composing (or decomposing) a larger unit from (or into 10) units of a lower value.
- 学习目标：Use the standard algorithm to add or subtract decimals with multiple non-zero digits.
- 课：6.5.1 Using Decimals in a Shopping Context；6.5.2 Using Diagrams to Represent Addition and Subtraction (Optional)；6.5.3 Adding and Subtracting Decimals with Few Non-Zero Digits；6.5.4 Adding and Subtracting Decimals with Many Non-Zero Digits
- 海报要点：相同数位的单位相加减，满十进一、退一作十；竖式
- 本地化 / 安全：原书购物情境用美元，本海报改用人民币（元）。

B部分 小数的乘法（Section B: Multiplying Decimals）

- 学习目标：Calculate the product of two decimals and explain the solution method.
- 学习目标：Use an algorithm to calculate the product of whole numbers, and justify how to use that value to find the product of two decimals with the same significant digits.
- 课：6.5.5 Using Fractions to Multiply Decimals；6.5.6 Methods for Multiplying Decimals；6.5.7 Using Diagrams to Represent Multiplication；6.5.8 Calculating Products of Decimals
- 海报要点：用分数理解小数乘法；面积图；先按整数乘再定小数点

C部分 小数的除法（Section C: Dividing Decimals）

- 学习目标：Calculate the quotient of a whole-number or a decimal dividend and a decimal divisor, and explain the solution method.
- 学习目标：Use long division to divide a multi-digit whole number or a decimal by a whole number.
- 课：6.5.9 Using Base-Ten Diagrams to Divide (Optional)；6.5.10 Using Partial Quotients；6.5.11 Using Long Division；6.5.12 Dividing Numbers that Result in a Decimal；6.5.13 Dividing a Decimal by a Decimal
- 海报要点：部分商、长除法；除数是小数时扩大成整数

D部分 综合运用（Section D: Let’s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions. The final lesson in this section is optional because it offers additional oppor…
- 课：6.5.14 Solving Problems Involving Decimals；6.5.15 Making and Measuring Boxes (Optional)
- 海报要点：解决小数问题；做盒子并测量
- 本地化 / 安全：做盒子要用剪刀和美工刀，海报加了安全提示；尺寸用厘米。

第6单元 表达式和方程（Unit 6: Expressions and Equations）

A部分 一元方程（Section A: Equations in One Variable）

- 学习目标：Solve equations of the form x+p=q or px=q and explain the solution method.
- 学习目标：Understand that solving an equation with a variable means finding a value for the variable that makes the equation true, and use substitution to determine whether a number is a solution to the equation.
- 课：6.6.1 Tape Diagrams and Equations；6.6.2 Truth and Equations；6.6.3 Staying in Balance；6.6.4 Practice Solving Equations；6.6.5 Represent Situations with Equations；6.6.6 Percentages and Equations
- 海报要点：解 x + p = q 和 px = q；代入检验解；天平模型

B部分 相等与等价（Section B: Equal and Equivalent）

- 学习目标：Justify whether two expressions are “equivalent,” or equal to each other for every value of their variable.
- 学习目标：Use the distributive property to write equivalent algebraic expressions.
- 课：6.6.7 Write Expressions with Variables；6.6.8 Equal and Equivalent；6.6.9 The Distributive Property, Part 1；6.6.10 The Distributive Property, Part 2；6.6.11 The Distributive Property, Part 3 (Optional)
- 海报要点：等价表达式对所有取值都相等；分配律

C部分 含指数的表达式（Section C: Expressions with Exponents）

- 学习目标：Evaluate expressions with whole-number exponents at specific values of their variables.
- 学习目标：Justify whether numerical expressions involving whole-number exponents are equal by evaluating the expressions and performing operations in the conventional order.
- 课：6.6.12 Meaning of Exponents；6.6.13 Expressions with Exponents；6.6.14 Evaluating Expressions with Exponents；6.6.15 Equivalent Exponential Expressions
- 海报要点：指数表示重复相乘；运算顺序；代值计算

D部分 两个量之间的关系（Section D: Relationships Between Quantities）

- 学习目标：Create a table, graph, and equation to represent the relationship between two quantities.
- 学习目标：Identify the independent and dependent variable in a situation.
- 课：6.6.16 Two Related Quantities, Part 1；6.6.17 Two Related Quantities, Part 2；6.6.18 More Relationships (Optional)
- 海报要点：自变量、因变量；表格、图像、方程

E部分 综合运用（Section E: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：6.6.19 Tables, Equations, and Graphs, Oh My!
- 海报要点：用表格、方程、图像描述真实情境

第7单元 有理数（Unit 7: Rational Numbers）

A部分 负数和绝对值（Section A: Negative Numbers and Absolute Value）

- 学习目标：Interpret a rational number and the absolute value of a number in context.

- 学习目标：Plot rational numbers and their opposites on a number line; know that a number and its opposite have the same absolute value.
- 学习目标：Use words and symbols to compare rational numbers, where a rational number could also be the absolute value of a number.
- 课：6.7.1 Positive and Negative Numbers；6.7.2 Points on the Number Line；6.7.3 Comparing Positive and Negative Numbers；6.7.4 Ordering Rational Numbers；6.7.5 Using Negative Numbers to Make Sense of Contexts；6.7.6 Absolute Value of Numbers；6.7.7 Comparing Numbers and Distance from Zero
- 海报要点：温度、海拔中的正负数；数轴；相反数；绝对值是到 0 的距离

B部分 不等式（Section B: Inequalities）

- 学习目标：Determine whether a given value is a solution to a given inequality.
- 学习目标：Draw and label a number line diagram to represent the solutions to an inequality.
- 学习目标：Write an inequality statement to represent a constraint.
- 课：6.7.8 Writing and Graphing Inequalities；6.7.9 Solutions of Inequalities；6.7.10 Interpreting Inequalities
- 海报要点：写不等式表示限制条件；在数轴上表示解集

C部分 坐标平面（Section C: The Coordinate Plane）

- 学习目标：Understand signs of numbers in ordered pairs as indicating locations in quadrants of the coordinate plane; recognize that when two ordered pairs differ only by signs, the locations of the points are related by reflections across one or both axes.
- 学习目标：Use coordinates and absolute value to find distances between points with the same first coordinate or the same second coordinate.
- 学习目标：Working in all four quadrants, plot a point given its coordinates, or identify the coordinates of a given point in the coordinate plane.
- 课：6.7.11 Points in the Coordinate Plane；6.7.12 Constructing the Coordinate Plane；6.7.13 Interpreting Points in a Coordinate Plane；6.7.14 Distances in the Coordinate Plane；6.7.15 Shapes in the Coordinate Plane
- 海报要点：四个象限；坐标符号与对称；同横坐标或同纵坐标的两点距离

D部分 公因数和公倍数（Section D: Common Factors and Common Multiples）

- 学习目标：List the factors of a number, and identify common factors for two numbers in a real-world situation.
- 学习目标：List the multiples of a number, and identify common multiples for two numbers in a real-world situation.
- 课：6.7.16 Common Factors；6.7.17 Common Multiples；6.7.18 Using Common Multiples and Common Factors
- 海报要点：列因数、倍数，找公因数、公倍数解决实际问题

E部分 综合运用：坐标画图（Section E: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. The lesson in this section is optional because it offers additional opportunities to practice standards that are not a focus of the grade. As this is a short section followed by an End-of-Unit Asse…

- 课：6.7.19 Drawing in the Coordinate Plane (Optional)
- 海报要点：在坐标平面上按点连线画图

第8单元 数据集和分布（Unit 8: Data Sets and Distributions）

A部分 数据和统计问题（Section A: Data, Variability, and Statistical Questions）

- 学习目标：Comprehend and use the terms “numerical” and “categorical” to describe data sets.
- 学习目标：Justify whether a question is “statistical” based on whether variability is expected in the data that could be collected.
- 课：6.8.1 Got Data?；6.8.2 Statistical Questions
- 海报要点：数值数据、分类数据；统计问题预期数据有变化

B部分 点图和直方图（Section B: Dot Plots and Histograms）

- 学习目标：Describe a distribution represented by a dot plot, including informal observations about its center and spread.
- 学习目标：Interpret a histogram to answer statistical questions about a data set.
- 课：6.8.3 Representing Data Graphically；6.8.4 Dot Plots；6.8.5 Using Dot Plots to Answer Statistical Questions；6.8.6 Interpreting Histograms；6.8.7 Using Histograms to Answer Statistical Questions；6.8.8 Describing Distributions on Histograms
- 海报要点：用点图、直方图描述分布的中心和离散程度

C部分 平均数和 MAD（Section C: Measures of Center and Variability）

- 学习目标：Compare the means and mean absolute deviations of different distributions.
- 学习目标：Interpret the mean and mean absolute deviation (MAD) in the context of the data.
- 课：6.8.9 Mean；6.8.10 Finding and Interpreting the Mean as the Balance Point；6.8.11 Variability and MAD；6.8.12 Using Mean and MAD to Make Comparisons
- 海报要点：平均数是均分值和平衡点；平均绝对离差 MAD

D部分 中位数和四分位距（Section D: Median and IQR）

- 学习目标：Compare and contrast distributions that are represented with box plots.
- 学习目标：Interpret the median and interquartile range (IQR) in the context of the data.
- 课：6.8.13 Median；6.8.14 Comparing Mean and Median；6.8.15 Quartiles and Interquartile Range；6.8.16 Box Plots；6.8.17 Using Box Plots
- 海报要点：中位数、四分位数、四分位距 IQR，箱线图

E部分 综合运用（Section E: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. Because this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions. The lesson in this section is optional because it offers additional opport…
- 课：6.8.18 Using Data to Solve Problems (Optional)
- 海报要点：用数据展示比较两个数据集

第9单元 综合运用（Unit 9: Putting It All Together）

A部分 探索我们的世界（Section A: Exploring Our World）

- 学习目标：Choose a method for comparing numbers.

- 学习目标：Estimate quantities in a real-world situation, and explain the estimation strategy.
- 学习目标：Solve problems involving rates.
- 课：6.9.1 Fermi Problems；6.9.2 Energy Flow (Optional)；6.9.3 Making Paper (Optional)；6.9.4 If Our Class Were the World (Optional)
- 海报要点：比较数、估算真实世界的数量、解比率问题
- 本地化 / 安全：原书「造纸」课比较不同原料，本海报只画比较方法，不编造数据。

B部分 投票（Section B: Voting）

- 学习目标：Choose and justify which voting system seems the fairest for dealing with more than two choices.
- 学习目标：Critique a statement reporting the results of a vote.
- 学习目标：Interpret voting situations involving more than two choices through the use of ratios and percentages.
- 课：6.9.5 How Do We Choose? (Optional)；6.9.6 More than Two Choices (Optional)；6.9.7 Comparing Voting Systems (Optional)；6.9.8 Picking Representatives (Optional)；6.9.9 Designing Districts (Optional)
- 海报要点：比和百分数解读投票；比较不同投票方法是否公平

C部分 建立联系（Section C: Making Connections）

- 学习目标：Coordinate diagrams and expressions involving equivalent fractions.
- 学习目标：Interpret and create diagrams involving a rectangle decomposed into squares.
- 课：6.9.10 Rectangle Madness (Optional)；6.9.11 Rectangle Fractions (Optional)
- 海报要点：长方形分割成正方形，与等值分数联系起来

七年级（Grade 7）· 9 个单元 · 36 个 Section

第1单元 比例图（Unit 1: Scale Drawings）

A部分 按比例缩放的图形（Section A: Scaled Copies）

- 学习目标：Determine whether a figure is a scaled copy of another figure, by examining corresponding side lengths and angle measures.
- 学习目标：Draw a scaled copy of a figure using a given scale factor.
- 课：7.1.1 What Are Scaled Copies?；7.1.2 Corresponding Parts and Scale Factors；7.1.3 Making Scaled Copies；7.1.4 Scaled Relationships；7.1.5 The Size of the Scale Factor；7.1.6 Scaling and Area (Optional)
- 海报要点：对应边成同一比例、对应角相等；按比例因子画缩放图

B部分 比例图（Section B: Scale Drawings）

- 学习目标：Create a scale drawing given the actual measurements of the object or given another scale drawing at a different scale.
- 学习目标：Explain how to use scales and scale drawings to calculate actual distances and areas.
- 课：7.1.7 Scale Drawings；7.1.8 Scale Drawings and Maps (Optional)；7.1.9 Creating Scale Drawings；7.1.10 Changing Scales in Scale Drawings；7.1.11 Scales without Units；7.1.12 Units in Scale Drawings
- 海报要点：比例尺；由比例图算实际距离和面积；换比例尺重画

C部分 综合运用：画平面图（Section C: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. Since this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions. The lesson in this section is optional because it offers additional opportun…
- 课：7.1.13 Draw It to Scale (Optional)
- 海报要点：按比例画教室或房间平面图

第2单元 认识比例关系（Unit 2: Introducing Proportional Relationships）

A部分 用表格表示比例关系（Section A: Representing Proportional Relationships with Tables）

- 学习目标：Determine the constant of proportionality for a proportional relationship represented in a table, and interpret it in context.
- 学习目标：Explain how to calculate unknown values in a table that represents a proportional relationship.
- 课：7.2.1 One of These Things Is Not Like the Others；7.2.2 Introducing Proportional Relationships with Tables；7.2.3 More about Constant of Proportionality
- 海报要点：比例常数 k；用表格求未知值

B部分 用方程表示比例关系（Section B: Representing Proportional Relationships with Equations）

- 学习目标：Use an equation to solve problems involving a proportional relationship.
- 学习目标：Write an equation of the form y = kx to represent a proportional relationship, given a table or a description of the situation.

- 课：7.2.4 Proportional Relationships and Equations；7.2.5 Two Equations for Each Relationship；7.2.6 Writing Equations to Represent Relationships
- 海报要点：写成 y = kx，用方程解题

C部分 比较比例与非比例关系（Section C: Comparing Proportional and Nonproportional Relationships）

- 学习目标：Determine whether the values in a table could represent a proportional relationship.
- 学习目标：Use a table to determine whether an equation represents a proportional relationship.
- 课：7.2.7 Comparing Relationships with Tables；7.2.8 Comparing Relationships with Equations；7.2.9 Solving Problems about Proportional Relationships
- 海报要点：判断表格、方程是否表示比例关系

D部分 用图像表示比例关系（Section D: Representing Proportional Relationships with Graphs）

- 学习目标：Determine whether a given graph represents a proportional relationship.
- 学习目标：Identify the constant of proportionality from the graph of a proportional relationship.
- 学习目标：Interpret points on the graph of a proportional relationship.
- 课：7.2.10 Introducing Graphs of Proportional Relationships；7.2.11 Interpreting Graphs of Proportional Relationships；7.2.12 Using Graphs to Compare Relationships；7.2.13 Two Graphs for Each Relationship
- 海报要点：过原点的直线；点 (1, k)；解读图上的点

E部分 综合运用：节约用水（Section E: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions. The first lesson is optional because it provides an opportunity for additional …
- 课：7.2.14 Four Representations (Optional)；7.2.15 Using Water Efficiently
- 海报要点：四种表示法；用比例关系算用水量

第3单元 圆的测量（Unit 3: Measuring Circles）

A部分 圆的周长（Section A: Circumference of a Circle）

- 学习目标：Recognize that there are proportional relationships between the circumference, diameter, and radius of circles and express these relationships using equations.
- 学习目标：Solve problems involving circumference.
- 课：7.3.1 How Well Can You Measure?；7.3.2 Exploring Circles；7.3.3 Exploring Circumference；7.3.4 Applying Circumference；7.3.5 Circumference and Wheels (Optional)
- 海报要点：周长 ÷ 直径 = π；C = πd = 2πr

B部分 圆的面积（Section B: Area of a Circle）

- 学习目标：Justify that the area of a circle can be calculated with the formula $A = \pi r^2$.
- 学习目标：Recognize that the area of a circle is not proportional to its diameter or radius.
- 学习目标：Solve problems involving the area of a circle.
- 课：7.3.6 Estimating Areas；7.3.7 Exploring the Area of a Circle；7.3.8 Relating Area to Circumference；7.3.9 Applying Area of Circles
- 海报要点：把圆剪拼成近似平行四边形；$A = \pi r^2$；面积与半径不成比例

C部分 综合运用：周长与面积（Section C: Let’s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions. The last lesson in this section is optional because it offers additional opport…
- 课：7.3.10 Distinguishing Circumference and Area；7.3.11 Stained-Glass Windows (Optional)
- 海报要点：区分周长和面积；设计彩色玻璃窗

第4单元 比例关系和百分数（Unit 4: Proportional Relationships and Percentages）

A部分 含分数的比例关系（Section A: Proportional Relationships with Fractions）

- 学习目标：Create algebraic expressions that represent a situation involving adding or subtracting a fraction of the initial value.
- 学习目标：Solve problems about proportional relationships with fractional quantities.
- 学习目标：Use long division to generate a decimal representation of a fraction.
- 课：7.4.1 Lots of Flags；7.4.2 Ratios and Rates with Fractions；7.4.3 Revisiting Proportional Relationships；7.4.4 More than That, Less than That；7.4.5 Say It with Decimals
- 海报要点：分数的比和比率；增加或减少原值的几分之几；长除法化小数
- 本地化 / 安全：原书第 1 课以美国国旗为情境，本海报只保留「按比例放大缩小」的数学，不使用国旗图案。

B部分 百分数的增加和减少（Section B: Percent Increase and Decrease）

- 学习目标：Create algebraic expressions or equations that represent a situation involving percent increase or decrease.
- 学习目标：Use diagrams to solve problems involving percent increase or decrease.
- 课：7.4.6 Increasing and Decreasing；7.4.7 One Hundred Percent；7.4.8 Percent Increase and Decrease with Equations；7.4.9 Part of a Percent
- 海报要点：增加 p% → 乘 (1 + p%)；用图和方程表示

C部分 百分数的应用（Section C: Applying Percentages）

- 学习目标：Calculate measurement error, and express it as a percentage of the actual value.
- 学习目标：Solve problems involving tax, tip, simple interest, markup, markdown, or commission.
- 课：7.4.10 Tax and Tip；7.4.11 Percentage Contexts；7.4.12 Solving Multi-step Percentage Problems；7.4.13 Measurement Error；7.4.14 Percent Error
- 海报要点：税、服务费、单利、加价、打折、提成；测量误差和百分误差
- 本地化 / 安全：原书的销售税（sales tax）和小费（tip）是美国情境；我国商品标价一般已含税、通常不付小费，本海报改用「服务费」「折扣」等情境，金额用人民币。

D部分 综合运用（Section D: Let’s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：7.4.15 Changes on the Earth (Optional)；7.4.16 Posing Percentage Problems
- 海报要点：用百分数理解地球上的变化；自己提出百分数问题

第5单元 有理数运算（Unit 5: Rational Number Arithmetic）

A部分 有理数的加减（Section A: Adding and Subtracting Rational Numbers）

- 学习目标：Apply addition and subtraction of signed numbers to represent situations and solve problems.
- 学习目标：Calculate the sum or difference of two rational numbers.
- 课：7.5.1 Interpreting Negative Numbers；7.5.2 Changing Temperatures；7.5.3 Changing Elevation；7.5.4 Money and Debts；7.5.5 Representing Subtraction；7.5.6 Finding Differences；7.5.7 Adding and Subtracting to Solve Problems
- 海报要点：温度、海拔、收支中的正负数加减；减法 = 加相反数

B部分 有理数的乘除（Section B: Multiplying and Dividing Rational Numbers）

- 学习目标：Apply multiplication and division of signed numbers to represent situations and solve problems.
- 学习目标：Calculate the product or quotient of two rational numbers.
- 课：7.5.8 Multiplying Rational Numbers (Part 1)；7.5.9 Multiplying Rational Numbers (Part 2)；7.5.10 Multiply! (Optional)；7.5.11 Dividing Rational Numbers；7.5.12 Negative Rates
- 海报要点：用运动方向和速度理解正负数相乘；除法是乘法的逆运算

C部分 有理数的四则运算（Section C: Four Operations with Rational Numbers）

- 学习目标：Apply the four operations with rational numbers to solve problems.
- 学习目标：Solve an equation of the form x + p = q or px = q, where p, q, and x are rational numbers.
- 学习目标：Write an equation of the form x + p = q or px = q (where p, q, and x are rational numbers) to represent a situation.
- 课：7.5.13 Expressions with Rational Numbers；7.5.14 Solving Problems with Rational Numbers；7.5.15 Solving Equations with Rational Numbers；7.5.16 Representing Contexts with Equations
- 海报要点：四则运算；解 x + p = q、px = q（p、q 可为负数）

D部分 综合运用：股票市场（Section D: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：7.5.17 The Stock Market
- 海报要点：用正负数描述股价涨跌
- 本地化 / 安全：原书股价用美元，本海报改用人民币，涨跌数字是示例。

第6单元 表达式方程和不等式（Unit 6: Expressions, Equations, and Inequalities）

A部分 表示两类情境（Section A: Representing Situations of the Form px+q=r and p(x+q)=r）

- 学习目标：Create diagrams and equations in the form px+q=r and p(x+q)=r to represent situations.
- 学习目标：Interpret equations in the form px+q=r and p(x+q)=r that represent relationships in diagrams and situations.
- 课：7.6.1 Relationships between Quantities；7.6.2 Reasoning about Contexts with Tape Diagrams；7.6.3 Reasoning about Equations with Tape Diagrams；7.6.4 Reasoning about Equations and Tape Diagrams (Part 1)；7.6.5 Reasoning about Equations and Tape Diagrams (Part 2)；7.6.6 Distinguishing between Two Types of Situations

- 海报要点：用线段图和方程表示 px + q = r 与 p(x + q) = r

B部分 解两类方程（Section B: Solving Equations of the Form px+q=r and p(x+q)=r and Problems That Lead to Those Equations）

- 学习目标：Solve equations of the form px+q=r and p(x+q)=r, including those that involve fractions, decimals, and negative numbers, and explain the solution method.
- 课：7.6.7 Reasoning about Solving Equations (Part 1)；7.6.8 Reasoning about Solving Equations (Part 2)；7.6.9 Dealing with Negative Numbers；7.6.10 Different Options for Solving One Equation；7.6.11 Using Equations to Solve Problems；7.6.12 Solving Problems about Percent Increase or Decrease
- 海报要点：天平模型：两边同做一样的事；含分数、小数、负数

C部分 不等式（Section C: Inequalities）

- 学习目标：Draw and label a graph on a number line that represents all the solutions to an inequality.
- 学习目标：Solve an inequality of the form px+q > r or px+q < r and interpret the solution.
- 学习目标：Write an inequality of the form px+q > r or px+q < r to represent a situation with a constraint.
- 课：7.6.13 Reintroducing Inequalities；7.6.14 Finding Solutions to Inequalities in Context；7.6.15 Efficiently Solving Inequalities；7.6.16 Interpreting Inequalities；7.6.17 Modeling with Inequalities
- 海报要点：解 px + q > r 或 < r，在数轴上表示解集并解释

D部分 写等价表达式（Section D: Writing Equivalent Expressions）

- 学习目标：Apply properties of operations to write an expression with fewer terms that is equivalent to a given expression.
- 学习目标：Apply the distributive property to factor or expand an expression.
- 课：7.6.18 Subtraction in Equivalent Expressions；7.6.19 Expanding and Factoring；7.6.20 Combining Like Terms (Part 1)；7.6.21 Combining Like Terms (Part 2)
- 海报要点：减法变加相反数；展开和因式分解；合并同类项

E部分 综合运用（Section E: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：7.6.22 Applications of Expressions
- 海报要点：用表达式解决实际问题

第7单元 角、三角形和棱柱（Unit 7: Angles, Triangles, and Prisms）

A部分 角的关系（Section A: Angle Relationships）

- 学习目标：Solve multi-step problems involving complementary, supplementary, and vertical angles.
- 学习目标：Write an equation to represent the relationship between angles in a given diagram.
- 课：7.7.1 Relationships of Angles；7.7.2 Adjacent Angles；7.7.3 Nonadjacent Angles；7.7.4 Solving for Unknown Angles；7.7.5 Using Equations to Solve for Unknown Angles
- 海报要点：互余、互补、对顶角；列方程求未知角

B部分 按条件画多边形（Section B: Drawing Polygons with Given Conditions）

- 学习目标：Draw triangles with two given angle measures and one side length, one given angle measure and two side lengths, or three side lengths.
- 学习目标：Justify whether 3 measures of angles or sides determine a unique triangle or more than one triangle, or if no triangle is possible.
- 课：7.7.6 Building Polygons (Part 1)；7.7.7 Building Polygons (Part 2)；7.7.8 Triangles with 3 Common Measures；7.7.9 Drawing Triangles (Part 1)；7.7.10 Drawing Triangles (Part 2)
- 海报要点：给定三边、两角一边、两边一角画三角形；能否唯一确定

C部分 立体几何（Section C: Solid Geometry）

- 学习目标：Calculate the surface area and volume of a prism.
- 学习目标：Decide whether to calculate the surface area or volume of a prism to solve a problem in a real-world situation.
- 课：7.7.11 Slicing Solids；7.7.12 Volume of Right Prisms；7.7.13 Decomposing Bases for Area；7.7.14 Surface Area of Right Prisms；7.7.15 Distinguishing Volume and Surface Area；7.7.16 Applying Volume and Surface Area
- 海报要点：截面；直棱柱体积 = 底面积 × 高；表面积；判断该用哪个

D部分 综合运用：做三棱柱（Section D: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. Because this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：7.7.17 Building Prisms (Optional)
- 海报要点：从零开始做一个三棱柱
- 本地化 / 安全：做棱柱要用剪刀，海报加了安全提示。

第8单元 概率和抽样（Unit 8: Probability and Sampling）

A部分 单步事件的概率（Section A: Probabilities of Single-Step Events）

- 学习目标：Use the results from a repeated experiment to estimate the probability of an event, and justify the estimate.
- 学习目标：Use the sample space to determine the probability of an event, and express it as a fraction, decimal, or percentage.
- 课：7.8.1 Mystery Bags；7.8.2 Chance Experiments；7.8.3 What Are Probabilities?；7.8.4 Estimating Probabilities through Repeated Experiments；7.8.5 More Estimating Probabilities；7.8.6 Estimating Probabilities Using Simulation
- 海报要点：样本空间；概率用分数、小数、百分数表示；重复试验估计概率

B部分 多步事件的概率（Section B: Probabilities of Multi-step Events）

- 学习目标：Describe a multi-step experiment that could be used to simulate a compound event in a real-world situation, and justify that it represents the situation.
- 学习目标：Interpret or create a list, table, or tree diagram that represents the sample space of a compound event, and use the sample space to write the probability for an event.
- 课：7.8.7 Simulating Multi-step Experiments；7.8.8 Keeping Track of All Possible Outcomes；7.8.9 Multi-step Experiments；7.8.10 Designing Simulations
- 海报要点：列表、表格、树状图列样本空间；设计模拟

C部分 抽样（Section C: Sampling）

- 学习目标：Describe methods to obtain a random sample from a population, and explain why it is representative of the population.
- 学习目标：Explain why samples are necessary and describe a sample and population for a given statistical question.
- 课：7.8.11 Comparing Groups；7.8.12 Larger Populations；7.8.13 What Makes a Good Sample?；7.8.14 Sampling in a Fair Way
- 海报要点：为什么要抽样；总体与样本；随机抽样才有代表性

D部分 用样本推断（Section D: Using Samples）

- 学习目标：Determine whether two populations are likely to be meaningfully different by reasoning about center and spread.
- 学习目标：Use the proportion of a random sample that is within a certain category to make inferences about the population, and explain the reasoning.
- 课：7.8.15 Estimating Population Measures of Center；7.8.16 Estimating Population Proportions；7.8.17 More about Sampling Variability (Optional)；7.8.18 Comparing Populations Using Samples；7.8.19 Comparing Populations with Friends
- 海报要点：用样本比例估计总体；比较两个总体的中心和离散

E部分 综合运用：记忆测试（Section E: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions. The lesson in this section is optional because it offers additional opportuniti…
- 课：7.8.20 Memory Test (Optional)
- 海报要点：设计实验、收集数据、比较两组

第9单元 综合运用（Unit 9: Putting It All Together）

A部分 经营餐馆（Section A: Running a Restaurant）

- 学习目标：Apply proportional reasoning, algebraic expressions, and percentages to model aspects of running a restaurant.
- 学习目标：Justify numerical decisions involved in running a restaurant.
- 课：7.9.1 Cost of a Meal (Optional)；7.9.2 Costs of Running a Restaurant (Optional)；7.9.3 Restaurant Floor Plan (Optional)
- 海报要点：用比例、表达式、百分数算菜品成本和餐馆开支，设计平面图
- 本地化 / 安全：原书金额用美元，本海报改用人民币，数字是示例。

B部分 建立联系（Section B: Making Connections）

- 学习目标：Apply proportional reasoning, algebraic expressions, and percentages to model real-world situations.
- 学习目标：Choose an appropriate level of accuracy when reporting estimates of quantities.
- 课：7.9.4 How Crowded Is This Neighborhood? (Optional)；7.9.5 Fermi Problems (Optional)；7.9.6 More Expressions and Equations (Optional)；7.9.7 Measurement Error (Optional)；7.9.8 Deforestation at Scale (Optional)
- 海报要点：人口密度、费米问题、测量误差、森林面积变化，选择合适的精确度

C部分 设计 5 千米路线（Section C: Designing a Course）

- 学习目标：Calculate the distance of a path using the circumference and number of rotations of a trundle wheel.
- 学习目标：Compare and contrast methods for measuring long distances.
- 课：7.9.9 Measuring Long Distances over Uneven Terrain (Optional)；7.9.10 Building a Trundle Wheel (Optional)；7.9.11 Using a Trundle Wheel to Measure Distances (Optional)；7.9.12 Designing a 5K Course (Optional)
- 海报要点：用测距轮的周长和圈数算距离，比较测量长距离的方法
- 本地化 / 安全：5 千米路线在室外测量，海报加了交通和场地安全提示。

八年级（Grade 8）· 9 个单元 · 40 个 Section

第1单元 刚体变换和全等（Unit 1: Rigid Transformations and Congruence）

A部分 刚体变换（Section A: Rigid Transformations）

- 学习目标：Determine coordinates that represent the image of a polygon or line segment in the coordinate plane after a transformation.
- 学习目标：Draw and label the image of figures that result from translations, rotations, and reflections on a square or isometric grid.
- 学习目标：Explain the sequence of transformations that takes one figure to its image.
- 课：8.1.1 Moving in the Plane；8.1.2 Naming the Moves；8.1.3 Grid Moves；8.1.4 Making the Moves；8.1.5 Coordinate Moves；8.1.6 Describing Transformations
- 海报要点：平移、旋转、反射；在网格和坐标平面上画像；描述变换序列

B部分 刚体变换的性质（Section B: Properties of Rigid Transformations）

- 学习目标：Draw and label rigid transformations of lines and parallel lines and explain the relationship between the original and its image under the transformation.
- 学习目标：Identify a rigid transformation using a drawing of a figure and its image.
- 学习目标：Identify side lengths and angles that have equivalent measurements in composite shapes and explain why they are equivalent.
- 课：8.1.7 No Bending or Stretching；8.1.8 Rotation Patterns；8.1.9 Moves in Parallel；8.1.10 Composing Figures
- 海报要点：不弯不拉：长度、角度不变；直线变直线，平行线仍平行

C部分 全等（Section C: Congruence）

- 学习目标：Compare and contrast side lengths, angle measures, and other features of shapes using rigid transformations to explain why a shape is or is not congruent to another.
- 学习目标：Justify that two polygons on a grid are congruent using the definition of congruence in terms of rigid transformations.
- 课：8.1.11 What Is the Same?；8.1.12 Congruent Polygons；8.1.13 Congruence
- 海报要点：用刚体变换定义全等，判断两个图形是否全等

D部分 三角形的内角（Section D: Angles in a Triangle）

- 学习目标：Calculate angle measures using alternate interior, vertical, and supplementary angles to solve problems.
- 学习目标：Generalize that the sum of angles in a triangle is 180 degrees using rigid transformations or the congruence of alternate interior angles of parallel lines cut by a transversal.
- 课：8.1.14 Alternate Interior Angles；8.1.15 Adding the Angles in a Triangle；8.1.16 Parallel Lines and the Angles in a Triangle
- 海报要点：内错角相等；三角形内角和 180°

E部分 综合运用：旋转与密铺（Section E: Let's Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment,

there are no section goals or checkpoint questions.
- 课：8.1.17 Rotate and Tessellate
- 海报要点：用变换做复杂的图案

第2单元 位似、相似和斜率（Unit 2: Dilations, Similarity, and Introducing Slope）

A部分 位似（Section A: Dilations）
- 学习目标：Create a dilation of a figure given a scale factor and center of dilation.
- 学习目标：Describe a figure on a coordinate grid and its image under a dilation, using coordinates to refer to points.
- 学习目标：Identify the center, scale factor, and image of a dilation.
- 课：8.2.1 Projecting and Scaling；8.2.2 Circular Grid；8.2.3 Dilations with No Grid；8.2.4 Dilations on a Square Grid；8.2.5 More Dilations
- 海报要点：位似中心、比例因子；在坐标平面上做位似

B部分 相似（Section B: Similarity）
- 学习目标：Calculate unknown side lengths in similar triangles using the ratios of side lengths within the triangles and the scale factor between similar triangles.
- 学习目标：Justify that two triangles are similar by finding a sequence of transformations that takes one triangle to the other or by checking that two pairs of corresponding angles are congruent.
- 课：8.2.6 Similarity；8.2.7 Similar Polygons；8.2.8 Similar Triangles；8.2.9 Side Length Quotients in Similar Triangles
- 海报要点：变换序列定义相似；两组对应角相等的三角形相似；用比求边长

C部分 斜率（Section C: Slope）
- 学习目标：Comprehend the term “slope” to mean a number that tells how steep a line is.
- 学习目标：Create an equation relating the quotient of the vertical and horizontal side lengths of a slope triangle to the slope of a line and use it to justify whether a point (x,y) is on the line by verifying that the values of x and y satisfy the equation.
- 课：8.2.10 Meet Slope；8.2.11 Writing Equations for Lines；8.2.12 Using Equations for Lines
- 海报要点：斜率表示直线的陡峭程度；斜率三角形；判断点是否在直线上

D部分 综合运用：影子测高（Section D: Let's Put It to Work）
- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：8.2.13 The Shadow Knows
- 海报要点：用相似三角形和影长求物体高度
- 本地化 / 安全：测影长时不要直视太阳，海报加了安全提示。

第3单元 线性关系（Unit 3: Linear Relationships）

A部分 正比例关系（Section A: Proportional Relationships）
- 学习目标：Create an equation and a graph to represent proportional relationships, including an appropriate scale and axes.
- 学习目标：Interpret multiple representations of a proportional relationship in context.

- 课：8.3.1 Understanding Proportional Relationships；8.3.2 Graphs of Proportional Relationships；8.3.3 Representing Proportional Relationships；8.3.4 Comparing Proportional Relationships
- 海报要点：写方程、画图（选合适的刻度和坐标轴），比较正比例关系

B部分 表示线性关系（Section B: Representing Linear Relationships）

- 学习目标：Create an equation that represents a linear relationship.
- 学习目标：Create and compare graphs that represent linear relationships with the same rate of change but different initial values.
- 学习目标：Interpret the slope and y-intercept of the graph of a line in context.
- 课：8.3.5 Introduction to Linear Relationships；8.3.6 More Linear Relationships；8.3.7 Representations of Linear Relationships；8.3.8 Translating to y=mx+b
- 海报要点：初始值和变化率；y = mx + b；斜率和 y 轴截距的意义

C部分 求斜率（Section C: Finding Slopes）

- 学习目标：Create multiple representations of a linear relationship, including a graph, equation, and table.
- 学习目标：Interpret the slope of a non-increasing line in context.
- 课：8.3.9 Slopes Don't Have to Be Positive；8.3.10 Calculating Slope；8.3.11 Line Designs (Optional)；8.3.12 Equations of All Kinds of Lines
- 海报要点：负斜率；用两点算斜率；水平线、竖直线的方程

D部分 线性方程（Section D: Linear Equations）

- 学习目标：Determine pairs of values that satisfy or do not satisfy a linear relationship using an equation or graph.
- 课：8.3.13 Solutions to Linear Equations；8.3.14 More Solutions to Linear Equations
- 海报要点：满足二元一次方程的数对就是直线上的点

E部分 综合运用（Section E: Let's Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：8.3.15 Using Linear Relations to Solve Problems
- 海报要点：为真实情境写线性方程并解释解

第4单元 线性方程和方程组（Unit 4: Linear Equations and Linear Systems）

A部分 等价方程（Section A: Equivalent Equations）

- 学习目标：Write equivalent equations and describe the moves that are used.
- 学习目标：Write equivalent equations to solve linear equations in one variable.
- 课：8.4.1 Writing Equivalent Equations；8.4.2 Keeping the Equation Balanced；8.4.3 Balanced Moves；8.4.4 More Balanced Moves；8.4.5 Solving Any Linear Equation；8.4.6 Strategic Solving
- 海报要点：天平模型；同时对两边做相同运算，解任意一元一次方程

B部分 一元一次方程的解（Section B: Linear Equations in One Variable）

- 学习目标：Describe features of linear equations that have one solution, no solution, or many solutions.
- 学习目标：Interpret the solution of an equation in one variable in context.

- 课：8.4.7 All, Some, or No Solutions；8.4.8 How Many Solutions?；8.4.9 When Are They the Same?
- 海报要点：唯一解、无解、无数解；结合情境解释

C部分 二元一次方程组（Section C: Systems of Linear Equations）

- 学习目标：Categorize systems of equations, including systems with infinitely many or no solutions, and calculate the solution for a system using a variety of strategies.
- 学习目标：Comprehend that solving a system of equations means finding values of the variables that make both equations true at the same time.
- 学习目标：Create a system of equations that represents a situation and interpret the solution in context.
- 课：8.4.10 On or Off the Line?；8.4.11 On Both of the Lines；8.4.12 Systems of Equations；8.4.13 Solving Systems of Equations；8.4.14 Solving More Systems；8.4.15 Writing Systems of Equations
- 海报要点：两条直线的交点；代入法；无解或无数解的方程组

D部分 综合运用（Section D: Let’ s Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：8.4.16 Solving Problems with Systems of Equations
- 海报要点：用方程组解决较难的实际问题

第5单元 函数和体积（Unit 5: Functions and Volume）

A部分 输入和输出（Section A: Inputs and Outputs）

- 学习目标：Comprehend the structure of a function as having one and only one output for each allowable input.
- 课：8.5.1 Inputs and Outputs；8.5.2 Introduction to Functions
- 海报要点：函数：每个输入有且只有一个输出

B部分 表示和解读函数（Section B: Representing and Interpreting Functions）

- 学习目标：Draw the graph of a function that represents a context, and explain which quantity is a function of which.
- 学习目标：Interpret multiple representations of functions, including graphs, tables, and equations, and explain how to find information in each type of representation.
- 课：8.5.3 Equations for Functions；8.5.4 Tables, Equations, and Graphs of Functions；8.5.5 More Graphs of Functions；8.5.6 Even More Graphs of Functions；8.5.7 Connecting Representations of Functions
- 海报要点：表格、方程、图像、故事之间互相转换

C部分 线性函数和变化率（Section C: Linear Functions and Rates of Change）

- 学习目标：Calculate the different rates of change of a piecewise linear function using a graph, and interpret the rates of change in context.
- 学习目标：Comprehend that any linear function can be represented by an equation in the form y=mx+b, where m and b are the rate of change and initial value of the function, respectively.
- 课：8.5.8 Linear Functions；8.5.9 Linear Models；8.5.10 Piecewise Linear Functions
- 海报要点：y = mx + b；分段线性函数的变化率

D部分 圆柱和圆锥（Section D: Cylinders and Cones）

- 学习目标：Calculate the value of one dimension of a cylinder or cone, and explain the reasoning.
- 学习目标：Calculate the volume of a cylinder or cone.
- 课：8.5.11 Filling Containers；8.5.12 How Much Will Fit?；8.5.13 The Volume of a Cylinder；8.5.14 Finding Cylinder Dimensions；8.5.15 The Volume of a Cone；8.5.16 Finding Cone Dimensions
- 海报要点：V圆柱 = $\pi r^2 h$，V圆锥 = $\frac{1}{3}\pi r^2 h$；由体积求半径或高
- 本地化 / 安全：往容器里倒水的实验，海报加了「擦干洒出的水，防滑」提示。

E部分 尺寸变化与球（Section E: Dimensions and Spheres）

- 学习目标：Calculate the volume of a sphere.
- 学习目标：Solve problems involving cones, cylinders, and spheres.
- 课：8.5.17 Scaling One Dimension；8.5.18 Scaling Two Dimensions (Optional)；8.5.19 Estimating a Hemisphere；8.5.20 The Volume of a Sphere；8.5.21 Cylinders, Cones, and Spheres
- 海报要点：改变一个尺寸对体积的影响；V球 = $4/3\ \pi r^3$

F部分 综合运用（Section F: Let's Put It to Work）

- 说明（官方未列 Section Goals）：The lesson in this section is optional because it offers additional opportunities to practice standards that are not a focus of the grade.
- 课：8.5.22 Volume as a Function of . . . (Optional)
- 海报要点：比较不同容器的水位高度

第6单元 数据中的关联（Unit 6: Associations in Data）

A部分 散点图（Section A: Does This Predict That?）

- 学习目标：Create a scatter plot from a table of data, and describe the trend of the data.
- 学习目标：Interpret a point on a scatter plot in context.
- 课：8.6.1 Organizing Data；8.6.2 Plotting Data；8.6.3 What a Point in a Scatter Plot Means
- 海报要点：由表格画散点图，描述趋势，解读一个点的含义

B部分 数值数据的关联（Section B: Associations in Numerical Data）

- 学习目标：Describe the relationship between two variables using a line fit to data on a scatter plot.
- 学习目标：Draw a linear model to fit data in a scatter plot, and describe features of a line that fits data well.
- 学习目标：Interpret features of data on a scatter plot, including linear and non-linear association, outliers, slope of a linear model, and clustering.
- 课：8.6.4 Fitting a Line to Data；8.6.5 Describing Trends in Scatter Plots；8.6.6 The Slope of a Fitted Line；8.6.7 Observing More Patterns in Scatter Plots；8.6.8 Analyzing Bivariate Data
- 海报要点：拟合直线、正负关联、非线性、离群值、聚集

C部分 分类数据的关联（Section C: Associations in Categorical Data）

- 学习目标：Calculate relative frequencies, and describe associations between variables using a relative frequency table.
- 学习目标：Create a two-way table and a segmented bar graph that represent relative frequencies, and interpret the frequencies in context.
- 课：8.6.9 Looking for Associations；8.6.10 Using Data Displays to Find Associations
- 海报要点：双向表、相对频率、分段条形图

D部分 综合运用：30 秒挑战（Section D: Let's Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions. The lesson in this section is optional because it offers additional opportuniti…
- 课：8.6.11 Gone in 30 Seconds (Optional)
- 海报要点：收集并分析计时数据

第7单元 指数和科学记数法（Unit 7: Exponents and Scientific Notation）

A部分 指数法则（Section A: Exponent Rules）

- 学习目标：Use exponent rules to generate equivalent numerical expressions for powers of 10.
- 课：8.7.1 Exponent Review；8.7.2 Multiplying Powers of 10；8.7.3 Powers of Powers of 10；8.7.4 Dividing Powers of 10；8.7.5 Negative Exponents with Powers of 10
- 海报要点：10 的幂：相乘、乘方、相除、负指数

B部分 更多指数法则（Section B: More Exponent Rules）

- 学习目标：Use exponent rules to generate equivalent numerical expressions for expressions with different bases and bases other than 10.
- 课：8.7.6 What about Other Bases?；8.7.7 Practice with Rational Bases；8.7.8 Combining Bases
- 海报要点：底数不是 10、底数是分数、底数不同

C部分 很大和很小的数（Section C: Large and Small Numbers）

- 学习目标：Compare very large or very small quantities expressed as a multiple of a power of 10.
- 学习目标：Use exponent rules and powers of 10 to solve problems in context.
- 课：8.7.9 Describing Large and Small Numbers Using Powers of 10；8.7.10 Representing Large Numbers on the Number Line；8.7.11 Representing Small Numbers on the Number Line；8.7.12 Applications of Arithmetic with Powers of 10
- 海报要点：用 10 的幂表示并在数轴上比较大数、小数

D部分 科学记数法（Section D: Scientific Notation）

- 学习目标：Calculate with numbers in scientific notation and interpret them in context.
- 学习目标：Identify numbers written in scientific notation, including scientific notation that has been generated by technology.
- 课：8.7.13 Definition of Scientific Notation；8.7.14 Estimating with Scientific Notation；8.7.15 Adding and Subtracting with Scientific Notation
- 海报要点：$a \times 10^n$（$1 \leqslant a < 10$）；用科学记数法估算和计算

E部分 综合运用：手机与登月（Section E: Let's Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：8.7.16 Is a Smartphone Smart Enough to Go to the Moon?
- 海报要点：用科学记数法比较数字存储和计算能力

第8单元 勾股定理和无理数（Unit 8: Pythagorean Theorem and Irrational Numbers）

A部分 正方形的边长和面积（Section A: Side Lengths and Areas of Squares）

- 学习目标：Comprehend the term “irrational number” to mean a number that is not rational and that $\sqrt{2}$ is an example of an irrational number.
- 学习目标：Comprehend the term “square root of n” and the notation $\sqrt{n}$ to mean the side length of a square whose area is n square units.
- 学习目标：Use the square root symbol to represent solutions to equations of the form $x^2=n$ and represent the square root as a point on the number line.
- 课：8.8.1 The Areas of Squares；8.8.2 Side Lengths and Areas；8.8.3 Square Roots；8.8.4 Rational and Irrational Numbers；8.8.5 Square Roots on the Number Line；8.8.6 Reasoning about Square Roots
- 海报要点：$\sqrt{n}$ 是面积为 n 的正方形的边长；$\sqrt{2}$ 是无理数；数轴上的平方根

B部分 勾股定理（Section B: The Pythagorean Theorem）

- 学习目标：Calculate the distance between two points in the coordinate plane by using the Pythagorean Theorem.
- 学习目标：Explain an area-based algebraic proof of the Pythagorean Theorem.
- 学习目标：Use the Pythagorean Theorem to calculate unknown side lengths of right triangles and to solve problems within a context.
- 课：8.8.7 Finding Side Lengths of Triangles；8.8.8 A Proof of the Pythagorean Theorem；8.8.9 Finding Unknown Side Lengths；8.8.10 The Converse；8.8.11 Applications of the Pythagorean Theorem；8.8.12 More Applications of the Pythagorean Theorem；8.8.13 Finding Distances in the Coordinate Plane
- 海报要点：$a^2 + b^2 = c^2$；面积证明；逆定理；求坐标平面上两点距离

C部分 正方体的棱长和体积（Section C: Side Lengths and Volumes of Cubes）

- 学习目标：Coordinate representations of a cube root, including cube root notation, decimal representation, the edge length of a cube of given volume, and a point on the number line.
- 课：8.8.14 Edge Lengths and Volumes；8.8.15 Cube Roots
- 海报要点：立方根 $\sqrt[3]{n}$ 是体积为 n 的正方体的棱长

D部分 有理数和无理数的小数（Section D: Decimal Representation of Rational and Irrational Numbers）

- 学习目标：Represent rational numbers as equivalent decimals and fractions.
- 课：8.8.16 Decimal Representations of Rational Numbers；8.8.17 Infinite Decimal Expansions
- 海报要点：有理数是有限或循环小数；循环小数化分数

E部分 综合运用：屏幕尺寸（Section E: Let's Put It to Work）

- 说明（官方未列 Section Goals）：In this final section, students have the opportunity to apply their thinking from throughout the unit. As this is a short section followed by an End-of-Unit Assessment, there are no section goals or checkpoint questions.
- 课：8.8.18 When Is the Same Size Not the Same Size?
- 海报要点：对角线相同、宽高比不同，屏幕面积不同

第9单元 综合运用（Unit 9: Putting It All Together）

A部分 密铺（Section A: Tessellations）

- 学习目标：Create tessellations using one or more polygons.
- 学习目标：Justify (in words and other representations) what shapes can and can't tessellate the plane.
- 课：8.9.1 Tessellations of the Plane (Optional)；8.9.2 Regular Tessellations (Optional)；8.9.3 Tessellating Polygons (Optional)
- 海报要点：哪些多边形能密铺平面，为什么

B部分 预测气温（Section B: Predicting the Temperature）

- 学习目标：Create a mathematical model of bivariate data using a scatter plot.
- 学习目标：Describe the limitations of a mathematical model.
- 课：8.9.4 What Influences Temperature? (Optional)；8.9.5 Plotting the Temperature (Optional)；8.9.6 Using and Interpreting a Mathematical Model (Optional)
- 海报要点：用散点图建立模型，讨论模型的局限